Fundamental and technical signals for A-share stock selection

Investments 1005 | Group 9

Historical daily data: 2 Jan 2018 to 29 Apr 2026. The main implementation is a long-only, equal-weight Top-100 portfolio with targets refreshed every 20 market sessions.

Research question

Can accounting and technical information improve stock selection after trading costs within the same opportunity set? We compare a fundamental score, a combined score, Ridge regression and a shallow LightGBM model. A matched equal-weight portfolio isolates stock-selection performance from differences in the available stock pool.

Main findings

Ridge earns 395.3% net over the complete history, versus 161.5% for the benchmark. Its maximum drawdown is 33.0%, compared with 32.4%. Higher historical return therefore does not imply lower downside risk.

Portfolio	Net return	Max. drawdown	Net excess (pp)
Fundamental	258.8%	-30.3%	+33.1
Combined	325.2%	-32.6%	+48.4
Ridge	395.3%	-33.0%	+63.8
LightGBM	360.4%	-33.7%	+59.5
Benchmark	161.5%	-32.4%	+0.0

For 2021–2025, Ridge returns 195.7% net, versus 76.6% for the benchmark. In 2026 through 29 Apr 2026, the corresponding returns are 8.80% and 8.18%; cumulative daily net excess is +0.56 percentage points.

All returns in the table use the same cadence and cost assumptions. Absolute returns compound daily wealth; excess sums daily strategy-minus-benchmark returns. These different definitions must not be interpreted as the same terminal-wealth difference. The partial 2026 record and historical implementation assumptions limit the strength of any claim about future performance.

1. Economic rationale and factor definitions

Profitability, growth, valuation and leverage may help explain relative returns. Price trends, volatility and turnover may capture information diffusion, reversal or liquidity. These are economic hypotheses; a plausible explanation alone does not establish a profitable trading rule.

ID	Prepared input and investment hypothesis
F1	Four-report weighted ROE relative to its trailing 60-day mean: changing profitability.
F2	120-day change in the four-report ROA slope / trailing 120-day standard deviation.
F3	The same slope-change transformation applied to revenue YoY growth.
F5	120-day change in debt/equity / trailing 120-day standard deviation: financial risk.
F6	Latest visible parent equity / total market value, retaining negative equity.
F7	Visible parent-profit TTM / total market value, retaining losses.
F8	Latest visible yearly ROA level: profitability.
F9	Latest visible operating cash flow / revenue: cash generation.
F10	Negative latest visible debt/assets: lower leverage.
T1, T2	Adjusted-close returns over 20 and 60 sessions: trend or reversal.
T3	Trailing 20-session population standard deviation of daily returns: volatility.
T5	Trailing 20-session mean turnover: trading activity and liquidity.
T6	Adjusted close at t-20 / adjusted close at t-126 minus one: medium-term momentum.

The adopted annual models use the union F1, F2, F3, F5, T1, T2, T3, T5, F6, F7, F8, F9. The 14 candidates above also define the common complete-input stock pool. F10 and T6 need not enter an adopted forecast to affect that common pool. Factor directions and annual input lists are fixed in config.json; the full formulas, report selection and missing-value conventions are in data_dictionary.md.

2. Historical data and the common opportunity set

The prepared daily panel aligns market prices, volume, turnover, historical universe membership, ST status, listing age, industry, market capitalization and financial snapshots on a fixed historical security axis. Inactive and subsequently delisted instruments remain on that axis; membership is not determined by survival today.

Data characteristic	Recorded value
Market sessions / historical slots	2,018 / 6,000
Mean base-eligible names	4,167
Mean common complete-input names	3,688
First signal / first execution	11 Jul 2018 / 12 Jul 2018

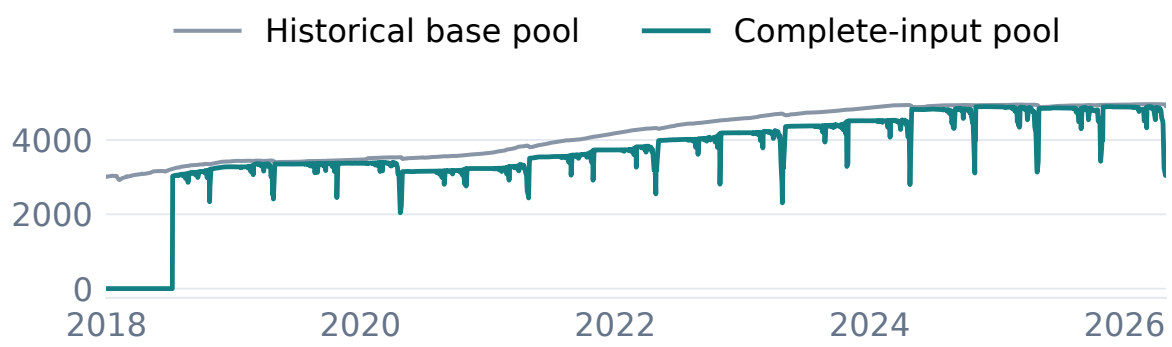


Figure 1. Daily base-eligible and common complete-input stock counts. These are available names, not actual portfolio holdings. Table means include every recorded session and the initial zero complete-input counts.

Eligibility requires historical Shanghai/Shenzhen A-share membership, non-ST status, at least six months listed, positive traded volume and valid market inputs. The common pool additionally requires all 14 candidate values. Every model, raw baseline and benchmark uses this same pool; future-label availability does not determine whether a stock can be selected.

Financial observations require report periods and both recorded announcement dates no later than the signal day. These guards reduce date inconsistencies but cannot certify the absence of later vendor revisions. Complete-input filtering also favors firms with a longer and more complete reporting history. Initial cash while lagged features accumulate explains the flat beginning before 12 Jul 2018.

3. Forecasts, score processing and fixed models

Each daily cross section is converted to centered average ranks. The prediction target is the adjusted opening return from t+1 to t+21, demeaned across eligible labeled stocks. Ridge provides a regularized linear comparison; a shallow tree model permits nonlinear interactions.

Method	Construction and fixed settings
Fundamental	Mean signed financial ranks. Score EMA span 20. Input union: F1, F2, F3, F6, F7, F8, F9.
Combined	Financial and technical family scores receive equal weight. Score EMA span 40. Input union: F1, F2, F3, F5, T1, T2, T3, T5.
Ridge	L2-regularized regression; alpha=10000; one BLAS thread. [1] Score EMA span 20. Input union: F1, F2, F3, F5, T1, T2, T3, T5.
LightGBM	Depth 3; 7 leaves; 56 trees; minimum leaf 500; learning rate 0.05. [2] Score EMA span 40. Input union: F1, F2, F3, F5, T1, T2, T3, T5.

Causal score processing

Prediction ranks are residualized against industry intercepts and within-industry log market capitalization, then reranked. A causal score EMA follows, with alpha=2/(span+1), followed by daily reranking. First finite values initialize state; missing inputs emit no score and preserve the previous state. State continues across year boundaries. Input EMA and the extra fundamental-score blend are zero.

Training chronology

Annual refits use expanding prior history sampled every 5 market sessions. A training observation is admitted only when its t+21 exit is strictly earlier than the prediction start. The 2018–2020 portion is a fitted replay using the pre-2021 model; from 2021, each annual model uses mature earlier labels. The full curve is therefore a historical backtest, not a claim that every point was an independent forecast.

Annual input lists can differ from their displayed unions. Model and processing choices used daily 2021–2025 development comparisons; the displayed portfolios replay the adopted 20-session implementation. Score residualization does not guarantee industry-neutral holdings or reduced drawdown.

4. Portfolio, benchmark and transaction accounting

The main portfolio selects the highest-scoring 100 names at equal target weights. Targets refresh every 20 market sessions on one continuous calendar. Scores observed after the preceding close determine next-open orders. Data, score updates and valuation remain daily, and blocked orders can be retried between target refreshes. Actual positions can differ from the intended Top 100.

The primary benchmark equal-weights the identical historical common pool, with the same target cadence, opening execution constraints and cost assumptions. It is an opportunity-set benchmark, not the CSI500 price index. Gross strategies are compared with the gross benchmark; net strategies with the net benchmark.

Cost or execution item	Assumption
Commission, each side	3 basis points
Slippage, each side	5 basis points
Sell stamp duty	10 bp before 28 Aug 2023; 5 bp thereafter [3, 4]
Target / execution / valuation	20-session refresh / next open / daily

Suspensions, zero volume and opening price limits can block trades. Cash limits and T+1 apply; an unsold holding remains in the book. Gross and net portfolios are simulated separately, so fees can also alter later quantities and cash. Round lots, minimum commissions, small transfer fees and liquidity impact are omitted. Daily bars do not establish auction queue priority.

Corporate actions and missing prices

A causal close/previous-close chain approximates reinvested corporate actions. Sourced share exchanges and cash exits are processed with their settlement assumptions; unavailable claims remain untradeable until release. Quote-less suspensions carry the last valid mark. The 60-day missing-price threshold is diagnostic, not an automatic exit. Unresolved formal listing ends assume zero recovery, which is a valuation convention rather than proof of zero eventual recovery.

The price chain does not reproduce exact cash-dividend receipt dates, dividend taxes or every rights subscription. These simplifications and uncertain historical data vintages affect the economic interpretation of simulated returns.

5. Complete-history absolute returns

Absolute curves show portfolio NAV minus one, using the complete daily history. They retain cash, execution constraints and compounding along each path.

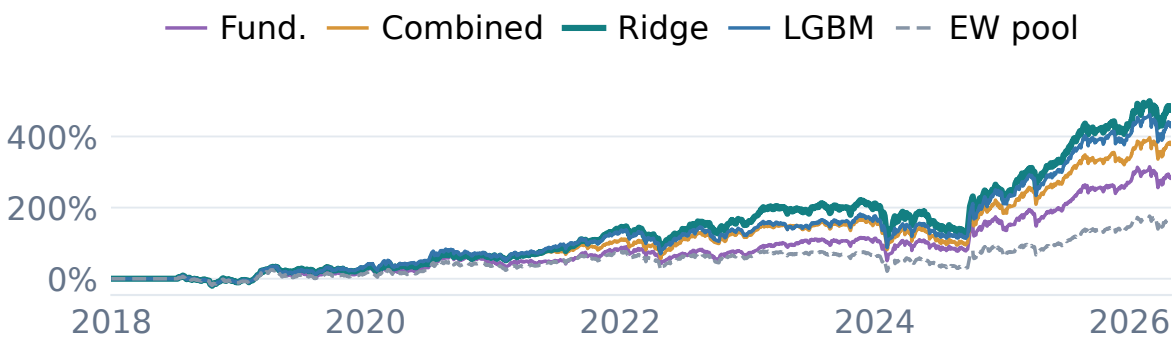


Figure 2. Cumulative gross returns, 2 Jan 2018–29 Apr 2026. Each strategy is compared with the gross common-pool benchmark.

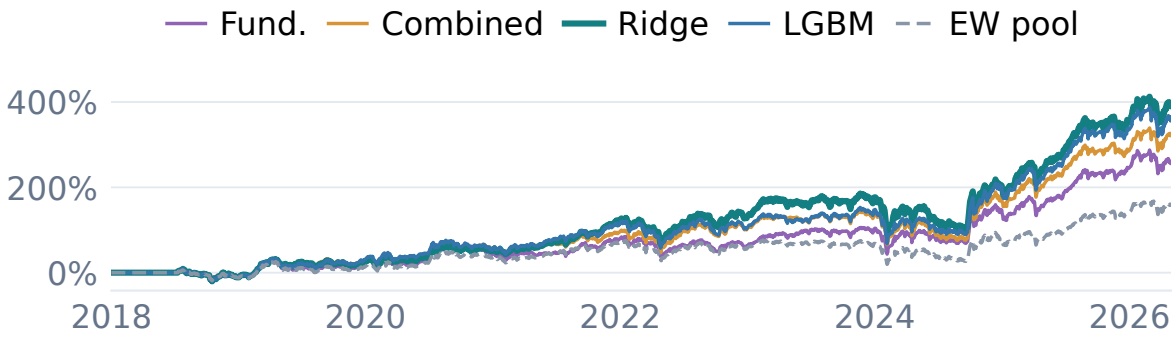


Figure 3. Cumulative net returns over the same complete history, including commission, sell stamp duty and slippage.

Ridge ends at 480.9% gross and 395.3% net. The benchmark ends at 165.5% gross and 161.5% net. The difference between a portfolio's gross and net terminal wealth reflects compounding and separately simulated trading paths; it is not the simple sum of its fee payments.

6. Complete-history excess returns

Each excess curve accumulates daily strategy returns minus the matching benchmark return. Units are percentage points. The benchmark is zero by construction; the measure does not compound a relative-wealth portfolio.

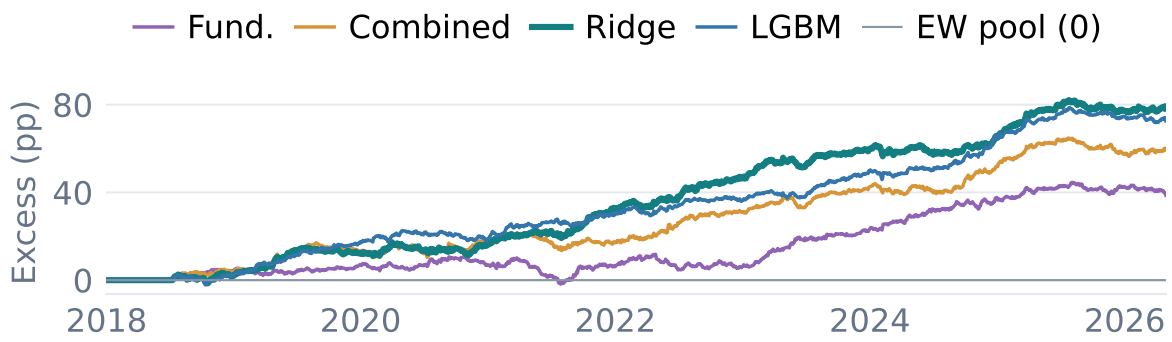


Figure 4. Cumulative daily gross excess versus the same-pool gross benchmark, 2 Jan 2018–29 Apr 2026.

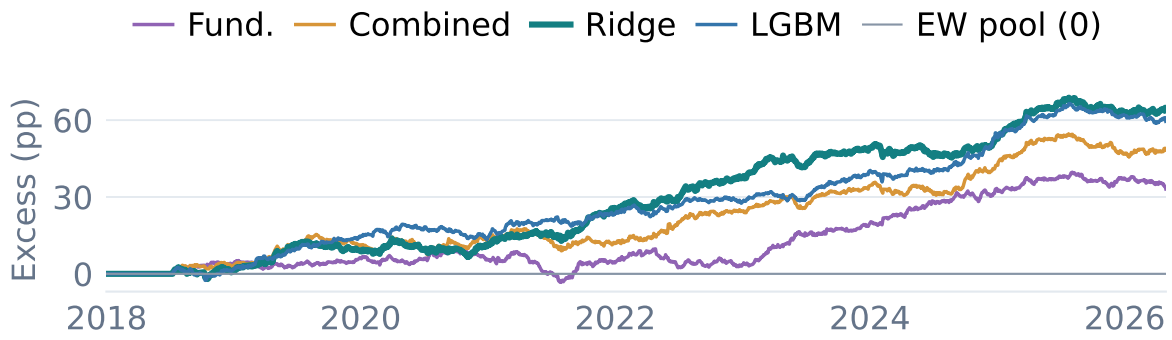


Figure 5. Cumulative daily net excess versus the same-pool net benchmark, including the matching execution costs.

Ridge accumulates +63.82 pp of daily net excess over the full history, including +52.02 pp in 2021–2025 and +0.56 pp in 2026 through 29 Apr 2026. A rising segment indicates positive cumulative selection advantage over that interval; a falling segment shows benchmark outperformance.

7. Drawdowns and the return–risk trade-off

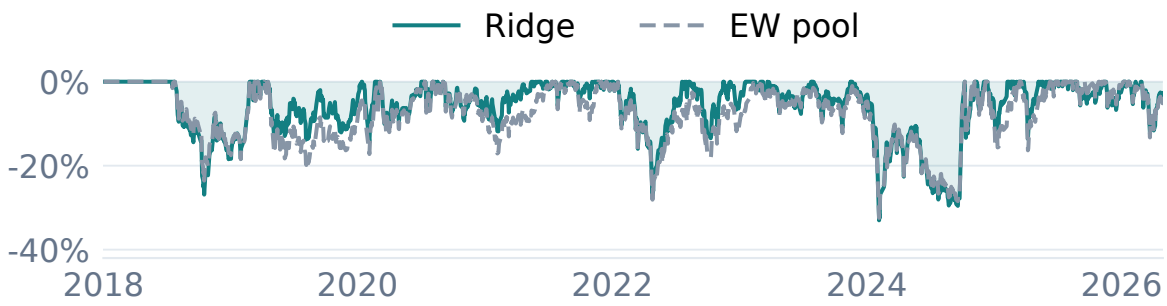


Figure 6. Net-wealth drawdowns relative to each portfolio’s prior peak. Negative values measure losses from that peak.

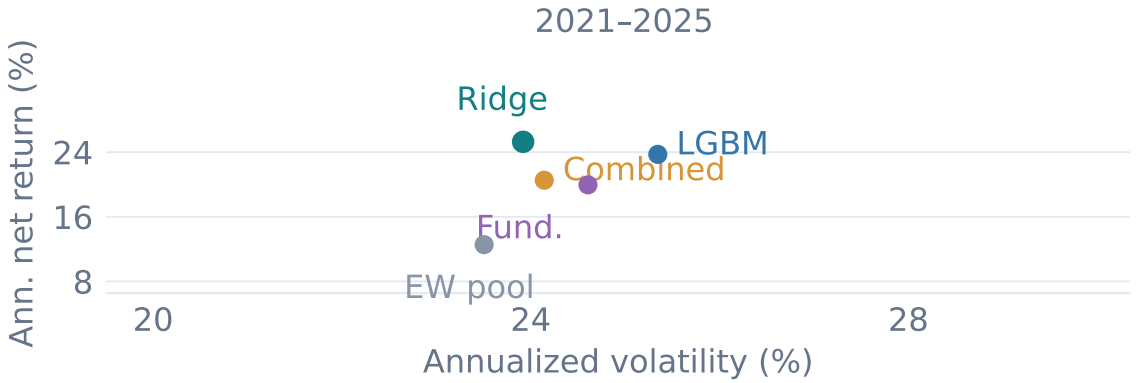


Figure 7. Net CAGR versus annualized daily volatility in 2021–2025. The table below instead records the complete active trading history.

Portfolio	Net CAGR	Annual vol.	Max. drawdown
Fundamental	18.6%	24.8%	-30.3%
Combined	21.3%	23.8%	-32.6%
Ridge	23.8%	23.9%	-33.0%
LightGBM	22.6%	25.5%	-33.7%
Benchmark	13.7%	23.9%	-32.4%

Ridge’s worst drawdown is 33.0%, slightly deeper than the benchmark’s 32.4%. Its annualized volatility is 23.9%, compared with 23.9%. The historical return advantage does not remove market exposure, drawdowns or model risk. Annualization uses 252 market sessions; no risk-free return is assumed for Sharpe.

8. Performance across months and calendar years

Monthly excess and calendar-year returns expose variation hidden by a final cumulative number. The two views aggregate the same daily observations using different definitions.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2018	0.0	0.0	0.0	0.0	0.0	0.0	+2.2	-1.2	-0.6	-0.1	+1.9	-1.2
2019	+1.7	+0.7	+1.1	+3.0	+1.4	+2.6	+1.0	-1.2	-0.6	-0.1	-0.5	-1.0
2020	+0.1	+0.1	+4.5	-1.7	-1.6	-1.1	-0.7	+0.8	+0.5	-2.3	+2.5	+1.0
2021	+2.7	+1.3	-1.0	+1.4	-0.3	+0.3	-2.8	+2.9	+3.1	+4.0	0.0	+2.5
2022	+2.1	+0.4	-2.0	+1.9	+1.1	+0.4	+3.8	+2.4	+0.1	+0.6	+0.3	+1.5
2023	+2.1	+3.1	+1.6	+0.6	-1.1	-2.0	+4.7	+0.2	+0.1	+0.2	+1.4	+0.4
2024	+0.5	-1.6	-0.7	+2.3	+0.2	-3.7	+0.7	-0.1	+2.5	+0.2	+0.8	+2.8
2025	+3.7	+2.4	+3.2	+2.5	+0.1	+1.5	+1.5	-0.8	-2.6	+0.1	-0.4	-0.9
2026	+0.2	-0.3	+0.1	+0.5	—	—	—	—	—	—	—	—



Figure 8. Ridge monthly net excess in percentage points: the sum of daily strategy-minus-benchmark returns. Gray dashes denote unobserved months.

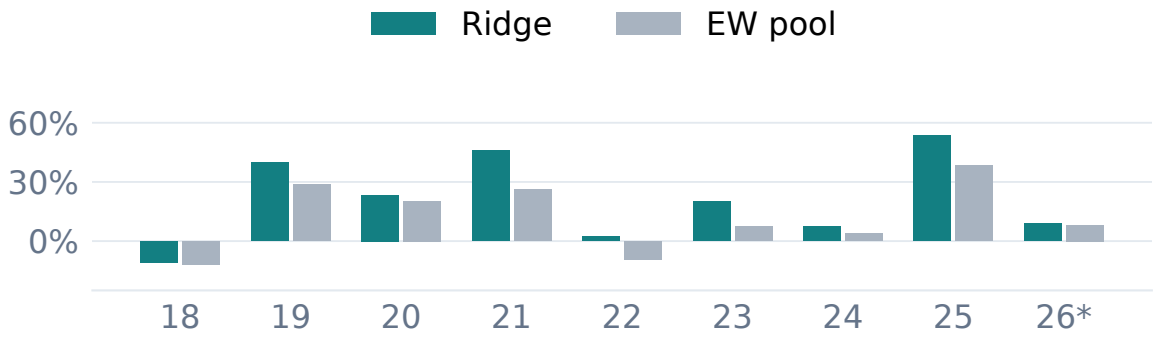


Figure 9. Compounded calendar-year net returns. The 2026 observations end on 29 Apr 2026; the final bar is a partial year.

The 2021–2025 Ridge net return is 195.7%, while 2026 through 29 Apr 2026 returns 8.80%. Negative monthly excess remains possible within a profitable overall history. The incomplete 2026 result is not directly comparable with a full-year bar and does not establish persistence across future market conditions.

9. Turnover and the sources of trading costs

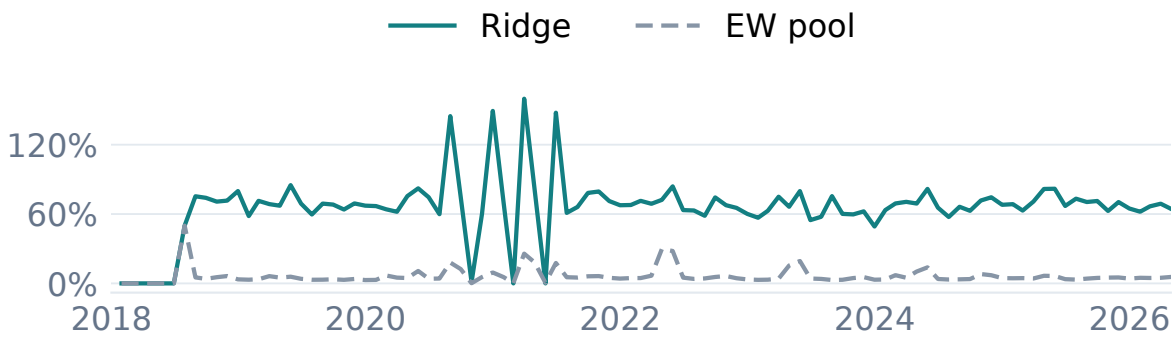


Figure 10. Monthly summed daily turnover for Ridge and the benchmark. Daily turnover is half the traded notional divided by opening wealth.

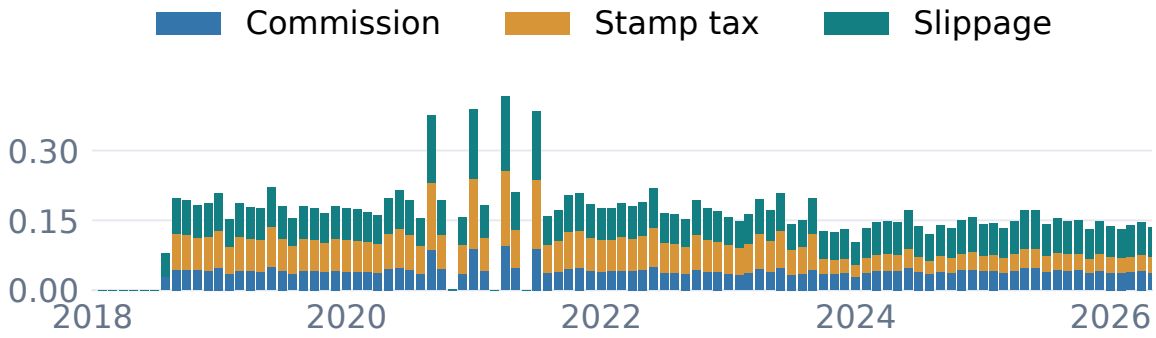


Figure 11. Ridge commission, stamp duty and slippage, scaled by each transaction day's pre-trade opening NAV and summed within month (pp).

Portfolio	Mean daily turnover	Annualized turnover
Ridge, 20-session targets	3.46%	8.71x
Benchmark, same targets	0.34%	0.84x
Ridge, daily targets	8.06%	20.31x

Twenty-session target updates reduce target replacement relative to the daily comparison, but blocked-order retries and changing portfolio values still create trades between updates. A 20-session strategy therefore need not trade on only one day per month. The component bars measure accounting charges; their sum must not be read as the gross-minus-net terminal return difference.

10. Input dependence and forecast rank correlation

Correlation identifies overlapping inputs; rank IC asks whether a higher model score tends to align with a higher future return across stocks. Neither measure includes turnover, transaction costs or actual opening fills.

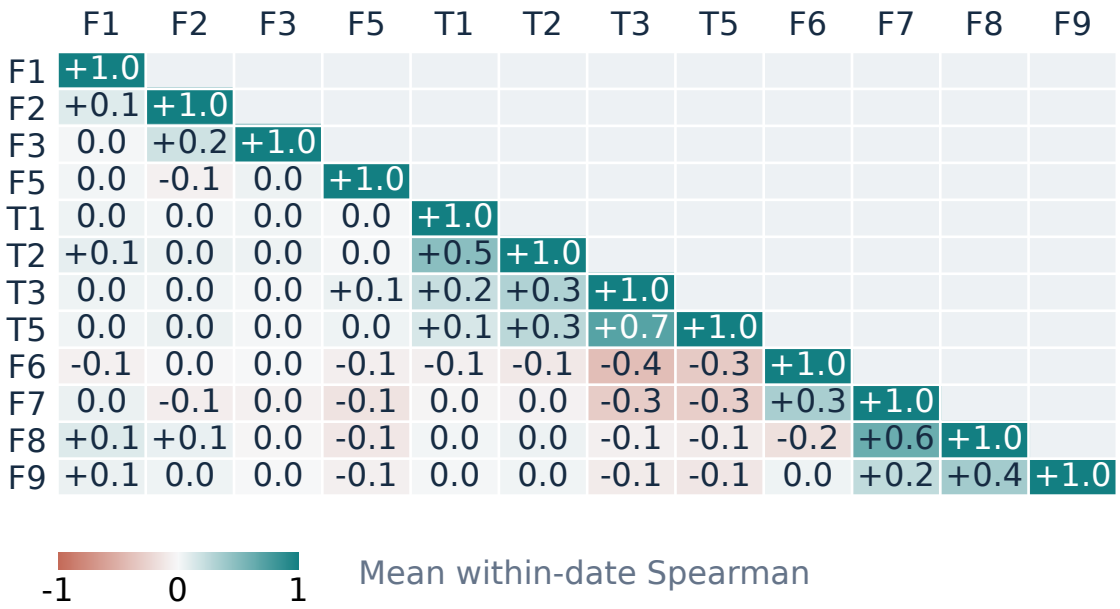


Figure 12. Adopted-input rank correlations: equal-date means of daily cross-sectional Spearman correlations in 2021–2025, sampled every fifth session in the common pool.

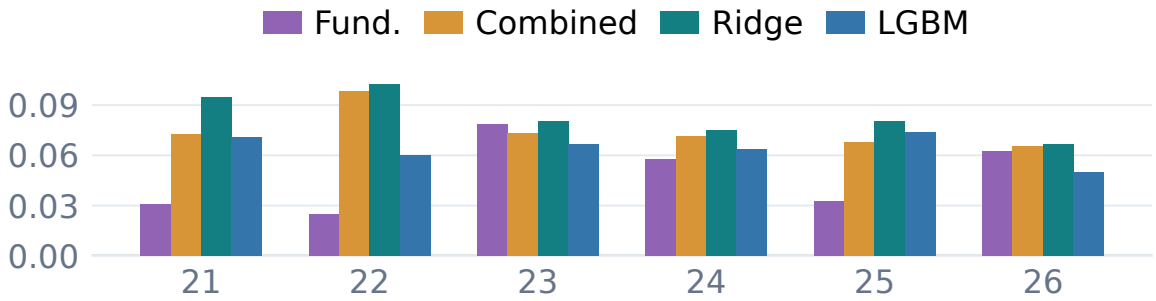


Figure 13. Annual mean daily Spearman rank IC of adopted scores against mature 20-session opening-return labels. The 2021–2025 values require label exits by 31 Dec 2025; 2026 exits end on 29 Apr 2026.

T3 and T5 have rank correlation +0.70, indicating overlapping volatility and activity information. Ridge's 2026 mean IC is +0.0665, but a positive whole-pool IC can coexist with weak high-score selection. The next page examines that tail. Overlapping labels make daily IC observations dependent, so the figure is descriptive.

11. Does the high-score group contain better returns?

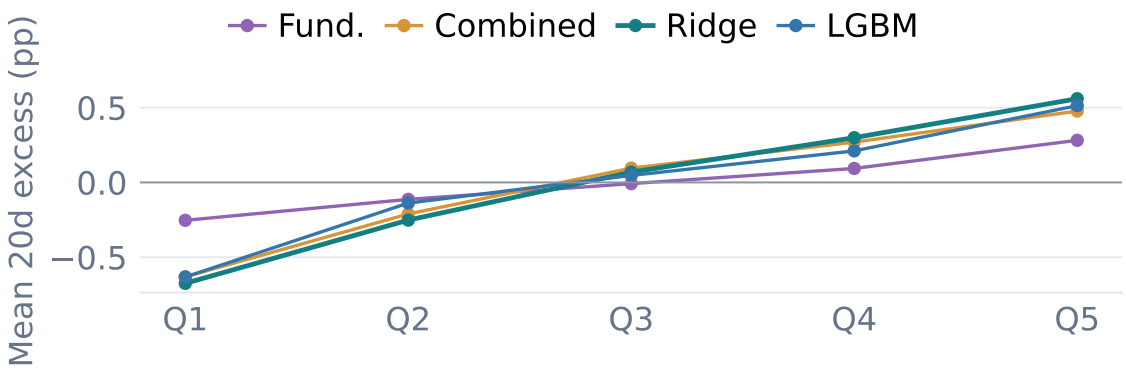


Figure 14. Mean future 20-session return excess by score quintile, 2021–2025. Q1 is the lowest score; Q5 is the highest.

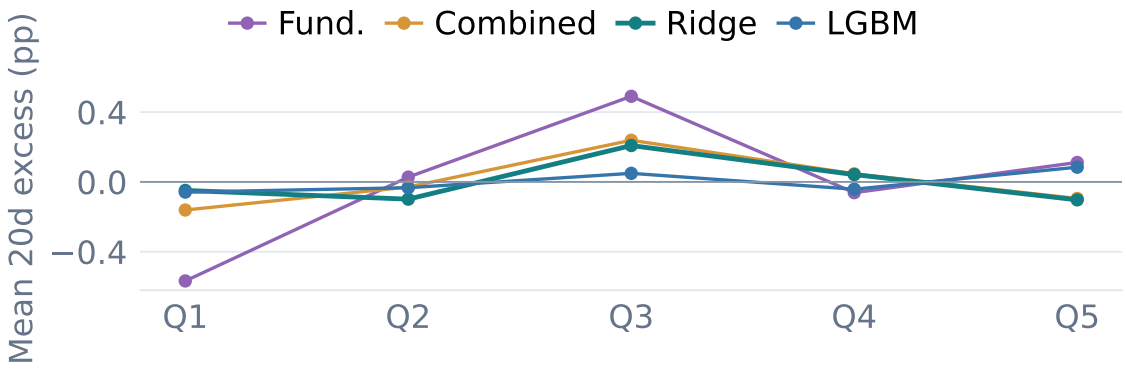


Figure 15. The same diagnostic for available 2026 signal dates through 30 Mar 2026. Later signals do not yet have mature labels.

Method	2021–2025 Q5-Q1 (pp)	2026 Q5-Q1 (pp)
Fundamental	+0.53	+0.68
Combined	+1.11	+0.07
Ridge	+1.23	-0.05
LightGBM	+1.15	+0.14

These are future-label cohort averages, not portfolio returns: they ignore fees, opening execution limits, cash and delayed sales. The comparison uses near-equal score groups before removing unavailable labels; ties use fixed security order. Excess subtracts the daily labeled score-pool mean, and daily cohort means receive equal weight. Labels ending after 2025 are excluded from the 2021–2025 panel. Adjacent cohorts overlap and are not independent. Ridge’s Q5-Q1 spread moves from +1.23 pp to -0.05 pp, weakening high-score separation in 2026.

12. Controls and the 2026 method comparison

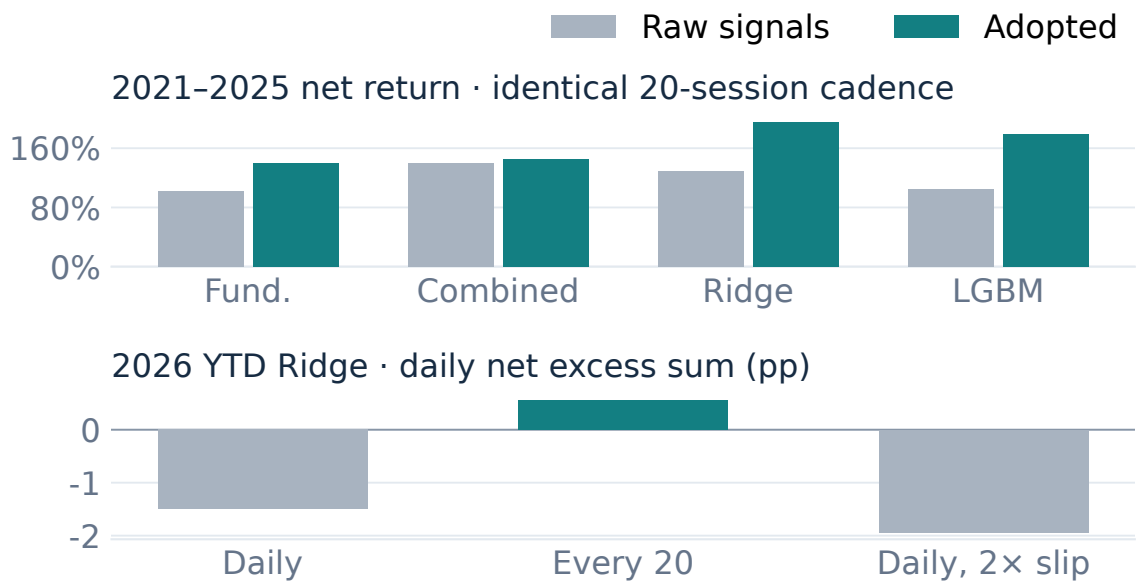


Figure 16. Above: 2021–2025 net returns for raw and adopted scores, with the same pool and 20-session schedule. Below: 2026 Ridge daily net excess under daily targets, 20-session targets and daily doubled slippage.

Portfolio	2026 gross	2026 net	Net excess (pp)
Fundamental	3.84%	3.53%	-4.25
Combined	10.21%	9.73%	+1.33
Ridge	9.40%	8.80%	+0.56
LightGBM	6.08%	5.56%	-2.40
Benchmark	8.23%	8.18%	+0.00

The raw 20-session Ridge baseline returns 128.6% net in 2021–2025, versus 195.7% for the adopted processed implementation. This comparison changes score processing while holding the historical pool and cadence fixed.

The doubled-slippage book is a daily implementation control, not a stress test of the main 20-session schedule. Comparisons across frequencies change both the timing of holdings and trading costs. The 2026 table retains every method. Combined leads this interval at 9.73% net, while Ridge remains the main model. The short record does not demonstrate a stable ordering of methods.

13. Limitations and conclusions

What the results support

The matched historical comparison supports further study of regularized cross-sectional ranking with causal score smoothing. Ridge combines a transparent linear model with a common opportunity set and explicit execution costs. Its stronger historical return is accompanied by meaningful drawdown and an inconsistent advantage across shorter intervals.

What remains uncertain

Historical specification choices can favor patterns that do not persist. The reviewed 2026 record is short and cannot establish an untouched prospective result. Overlapping return labels limit statistical independence. Complete-input filtering changes the opportunity set, while vendor revisions and announcement timestamps do not certify original unrevised accounting vintages.

Execution uses daily opening information and omits market impact, queue priority, lot rounding and minimum commissions. Corporate-action valuation and unresolved exit recovery are approximations. Score residualization removes specific cross-sectional components before ranking; it does not enforce neutral actual holdings or guarantee lower risk.

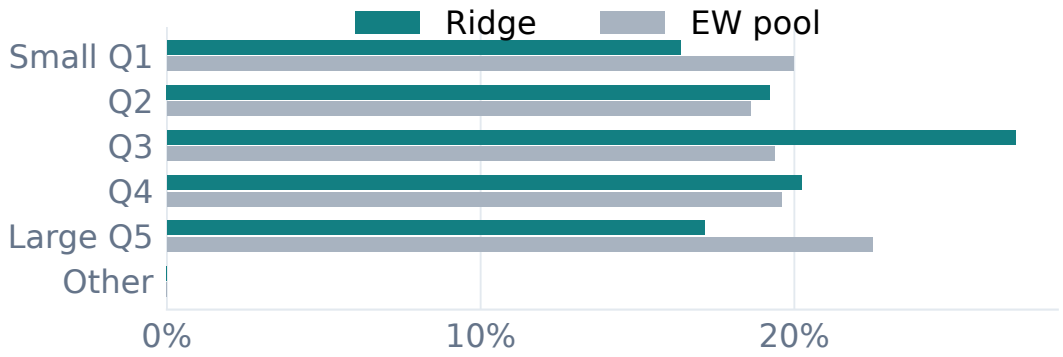


Figure 17. Actual net-NAV weights on 29 Apr 2026 by market-cap quintile (Q1 small, Q5 large), using same-day eligible-pool boundaries. Other combines cash, pending claims and unknown capitalization.

The main conclusion is a historically attractive net stock-selection result with material implementation and estimation limits. Ridge's 8.80% net return in 2026 through 29 Apr 2026, against 8.18% for the matched benchmark, is modest evidence over a partial year. A future prospective run with unchanged rules would provide a stronger check.

References and reproducibility

[1] Scikit-learn. *Ridge regression documentation.* Objective and L2-regularization parameter. Official API documentation.

[2] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T.-Y. (2017). *LightGBM: A Highly Efficient Gradient Boosting Decision Tree.* Advances in Neural Information Processing Systems 30. Conference paper.

[3] Ministry of Finance, China (2008). *Securities transaction stamp duty changed to sell-side collection from 19 September 2008.* Official announcement.

[4] Ministry of Finance and State Taxation Administration, China (2023). *Announcement No. 39: Halving securities transaction stamp duty from 28 August 2023.* Official announcement. Commission and slippage are research assumptions.

Reproduction from prepared external inputs

config.json records fixed annual factor selections, directions, model settings and trading assumptions. data_dictionary.md defines the prepared arrays, factor formulas, financial visibility guards and external corporate-action table. The calculation code retrains the fixed models and reconstructs all selected and raw-baseline portfolios from those inputs; no previous score or return file is required as a model input.

Generated external results contain the continuous daily ledger and metrics for the 20-session main implementation, daily comparison and daily doubled-slippage comparison. The raw baseline has its own matched 20-session ledger. Charts use those ledgers, while rank-IC and score-group figures use mature labels and adopted scores. Obtaining or rebuilding the supplier’s original financial snapshots is outside the prepared-input reproduction.

The report contains 14 main-body pages, including all figures and tables, plus this reference page. Body text, table text, chart labels and captions use 12-point type; body leading is 18 points with one-inch margins. Report figures are embedded as vector PDFs.